

≧ Lecture 1: Functions ≦

Math 1A / Calculus PREP

1) INTRODUCTION

Before we can talk about limits, derivatives, or integrals we need to understand the main object calculus works with: FUNCTIONS

A function describes how one quantity depends on one another

- The area of a circle depends on its radius
- Temperature depends on time
- Cost to mail a package depends on its weight

GOALS

By the end of today's lecture, you should be able to answer:

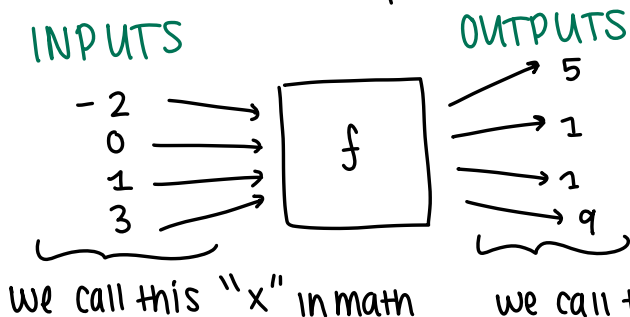
- ① What is a function?
- ② What is a domain and range?
- ③ How can we represent functions?

2. DEFINITION OF A FUNCTION

A function with domain D is a rule that assigns every input x in D exactly one output in $f(x)$.

3. FUNCTION MACHINE

think of a function like a machine that takes input applies a function and makes the outputs

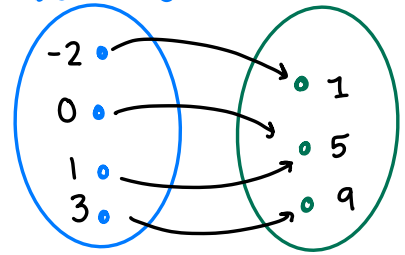


4. DOMAIN & RANGE

Domain = all possible inputs
(allowed x -values).

Range = all possible outputs $f(x)$ or y

DOMAINS



5. Real world examples :

WEATHER

Let $f(\text{day}) = \text{typical weather on that day of year}$

Inputs (x): Day of Year	Outputs ($f(x)$) Weather
day 1 (Jan 1) ❄️	cold / snowy
day 36 (Feb 5) ☁️	rainy
day 200 (July 19) ☀️	sunny

Domain: $\{1, 2, 3, \dots, 365\}$ (all days of year)

Range: $\{\text{cold/snowy, rainy, sunny, ...}\}$ (types of weather)

6 Important concept

Different inputs may share the same output. THAT'S OKAY. But one input cannot have 2 different outputs.

OKAY ✓

$$f(2) = 4$$

$$f(-2) = 4$$

Same output,
different inputs.

X NOT OKAY

$$f(1) = 3$$

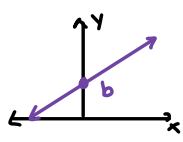
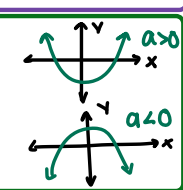
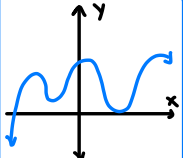
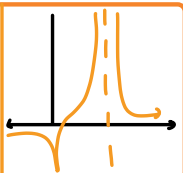
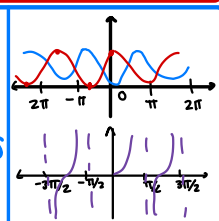
$$f(1) = 2$$

Same input
different outputs

1) FUNCTIONS IN MATH COMMON TYPES

Many functions in calculus come from a few families learning their shapes & formulas helps us recognize them.

↳ Note: You do not need to memorize these just recognize these families by formula & graph

TYPE	GENERAL FORM	KEY NOTES	GRAPH
LINEAR	$f(x) = mx + b$	• Straight line • m = slope • b = y intercept • degree = 1	
QUADRATIC	$f(x) = ax^2 + bx + c$ $\downarrow \quad \downarrow \quad \downarrow$ any number	• parabola • degree 2 • opens up $\curvearrowright a > 0$ • opens down $\curvearrowleft a < 0$	
Polynomials	$f(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$	• built from powers of x • no breaks • highest power = degree	
Rational	$f(x) = \frac{p(x)}{q(x)}$ where p & q are polynomials	• denominator cannot be 0 • may have asymptotes (vertical, horizontal, slant) • domain excludes 0 of $q(x)$	
Exponential	$f(x) = b^x$	• variable is in the exponent • domain: all real #'s ($\mathbb{R}$) • range: $(0, \infty)$ • growth: $b > 1$ • decay: $0 < b < 1$	
TRIG.	$\sin x$ $\cos x$ $\tan x$	• periodic behavior • oscillation for sin/cos • tan has repeating asymptotes (undefined @ $x = \frac{\pi}{2} + k\pi$)	

PRACTICE Identify the Type:

- (a) $f(x) = 5x - 2$ → line or
 (b) $g(x) = x^2 - 4x + 1$ → quadratic
 (c) $h(x) = \frac{x+1}{x-3}$ → rational
 (d) $p(x) = 2^x$ → exponential

How can we represent function?

A single function can be shown in different ways. Each representation gives us useful info.

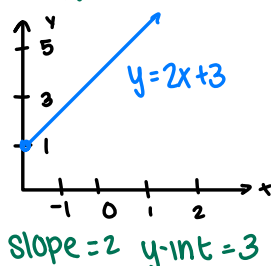
4 ways to represent a function?

FORMULA

$$f(x) = 2x + 3$$

This is the exact rule. It works for any input x in domain.

GRAPH



TABLE

x	y
-1	1
0	3
1	5
2	7

inputs outputs

WORDS

Multiply 2 then add 3

Describes the rule in words

FORMULA

- Shows the exact rule
- Easy to compute outputs for any input

GRAPHS

- Shows overall behavior & shape
- Helps us see trends: increasing, decreasing, and intercepts

TABLES

- Shows selected input - outputs value
- Great for looking for patterns

WORDS

- Explains the relationship in plain language
- math → real world

Practice

Given the rule: Each cookie costs \$2 plus \$3 service fee write it as a formula & make a table for 1, 2, 4 cookies

A) Formula = $C(x) = 2x + 3$

B)

Number of cookies (x)	1	2	4
Total cost C(x)	5	7	11

Reminder A relation is a function if every input has exactly one output

GRAPHS: VERTICAL LINE TEST

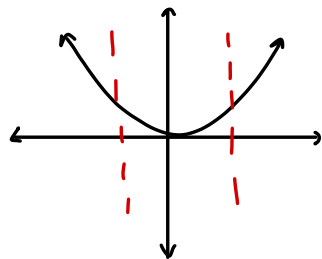
def: A graph is the graph of a function if every vertical line intersects it at most once

why does this work

A vertical line fixes one x value. If the graph is hit twice (or more), then that one input x has 2 different outputs. That breaks the definition of a function.

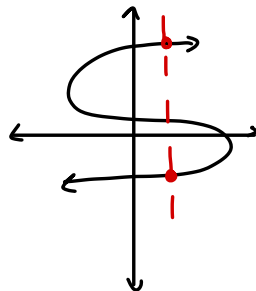
Ex:

✓ PASSES



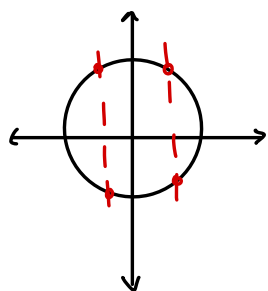
Any vertical line hits the graph at most once.

✗ FAILS



Some vertical lines hit the graph twice

✗ FAILS



Some vertical lines hit graph twice

TABLES HOW TO TEST A TABLE REPRESENT A FUNCTION

* Check whether the same input appears twice with different outputs *

✓ Passes

x	1	2	4	6
y	2	3	5	7

each input appears once ✓

✗ Fails

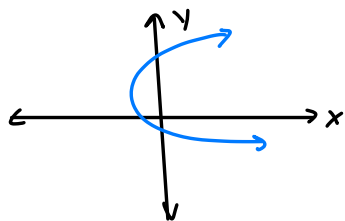
x	1	2	1	5
y	0.5	3	6.75	10

Input 1 appears twice with different outputs (0.5 and 6.75) This breaks defn.

Note: Repeated outputs are okay
Repeated inputs w/ different outputs IS NOT OKAY

PRACTICE

a) Is this a graph of a function



No! some vertical lines hit it twice

b) Is this a table of a function?

x	-1	0	2	2
y	4	5	6	6

Yes! b/c the repeated input 2 still has same output 6

Intercepts Graph

• **X intercept**: where graph crosses x-axis, so output $y=0$

• **y int**: where graph crosses y-axis, so $x=0$

COORDS

x intercepts $(x, 0)$

y intercepts $(0, y)$

Intercepts Formula

Example $f(x) = 2x + 10$

X intercepts (set $f(x) = 0$):

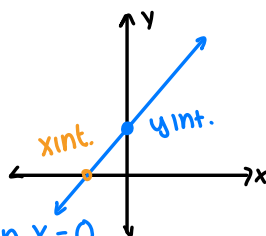
$$f(x) = 0$$

$$2x + 10 = 0$$

$$2x = -10$$

$$x = -5$$

Point: $(-5, 0)$



y intercepts plug in $x=0$

$$f(0) = 2(0) + 10$$

$$f(0) = 10$$

Point: $(0, 10)$

Intersections

Intersections: points where 2 graphs meet.

At an intersection, the outputs are equal

How to find algebraically

Ex: $f(x) = 2x + 10$ & $h(x) = 3x + 12$

Set functions equal:

$$2x + 10 = 3x + 12$$

$$-x = 2$$

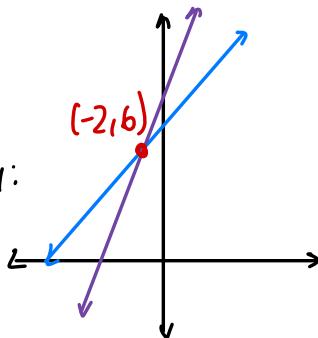
$$x = -2$$

Plug back in to find y:

$$f(-2) = 2(-2) + 10$$

$$= -4 + 10 = 6$$

Point: $(-2, 6)$



PRACTICE

Find the x & y intercepts of $y = x^2 - 4$

a) $\hookrightarrow (-2, 0)$ & $(2, 0)$ x int
 $\hookrightarrow (0, -4)$ y int

b) $y = x + 1$ $y = 2x - 2$ Intersection
 $x = 3$ $y = 4$ $(3, 4)$

How do we describe how function changes as we move left to right on its graph

Increasing & Decreasing

A function is increasing on an interval if whenever $a < b$ then $f(a) < f(b)$

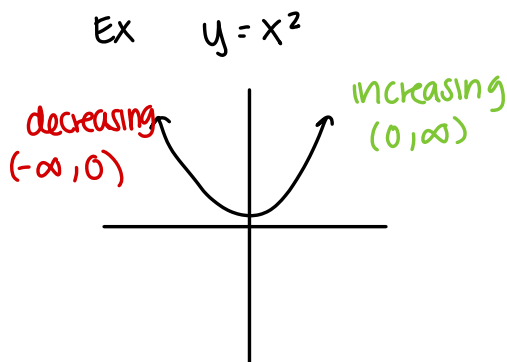
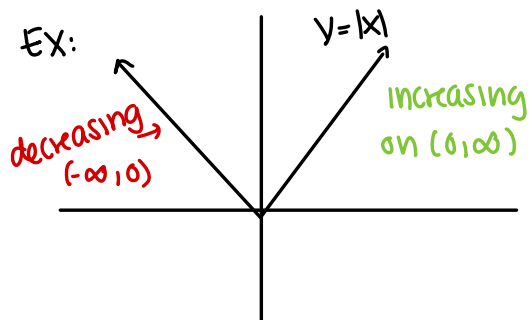
A function is decreasing on a interval if whenever $a < b$ $f(a) > f(b)$

↗ Increasing

graph goes up
as we move left to
right

↘ decreasing

graph goes
down as we
move left to
right



* a parabola opening up is decreasing until vertex then increasing after vertex *

Interval Notation Note:

We usually describe increasing/decreasing using intervals of x values

Ex:

- Increasing on $(0, \infty)$ means for all $x > 0$
- Decreasing on $(-\infty, 0)$ means f is decreasing for all $x < 0$

Practice:

a) for a line w/ pos slope

is it increasing or decreasing?

b) For a line w/ negative

slope is it increasing or decreasing?

Symmetry

① EVEN FUNCTIONS

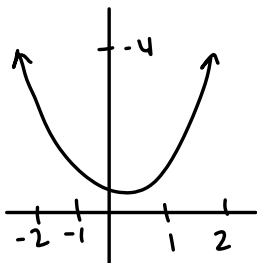
- Def: $f(-x) = f(x)$
- Graph Meaning: y axis symmetry

example:

$$f(x) = x^2$$

$$f(-2) = (-2)^2 = 4$$

$$f(2) = (2)^2 = 4$$



② ODD FUNCTIONS

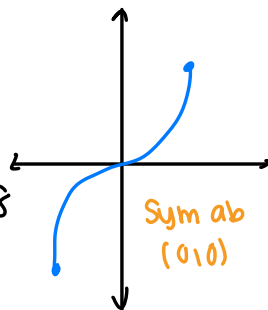
- Defn: $f(-x) = -f(x)$
- Graph: origin symmetry meaning (same after 180° rotation)

Example:

$$f(x) = x^3$$

$$f(2) = (2)^3 = 8$$

$$f(-2) = (-2)^3 = -8$$



Neither odd or even

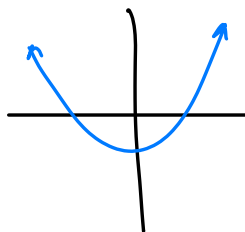
$$\text{EX: } f(x) = x^2 + x$$

$$f(3) = 3^2 + 3 = 9 + 3 = 12$$

$$f(-3) = (-3)^2 + (-3) = 9 - 3 = 6$$

Since $f(-3) = 6 \neq f(3) = 12$
it is NOT even

Since $f(-3) = 6 \neq -f(3) = -12$
it is not odd



PRACTICE

a) $f(x) = x^4$

even, odd or neither?

$$f(-x) = (-x)^4 = x^4 = f(x)$$

b) Is $f(x) = x^5$

odd, even, neither

$$f(-x) = (-x)^5 = -x^5 = -f(x) \text{ odd}$$